



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SIVAMBIGA FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

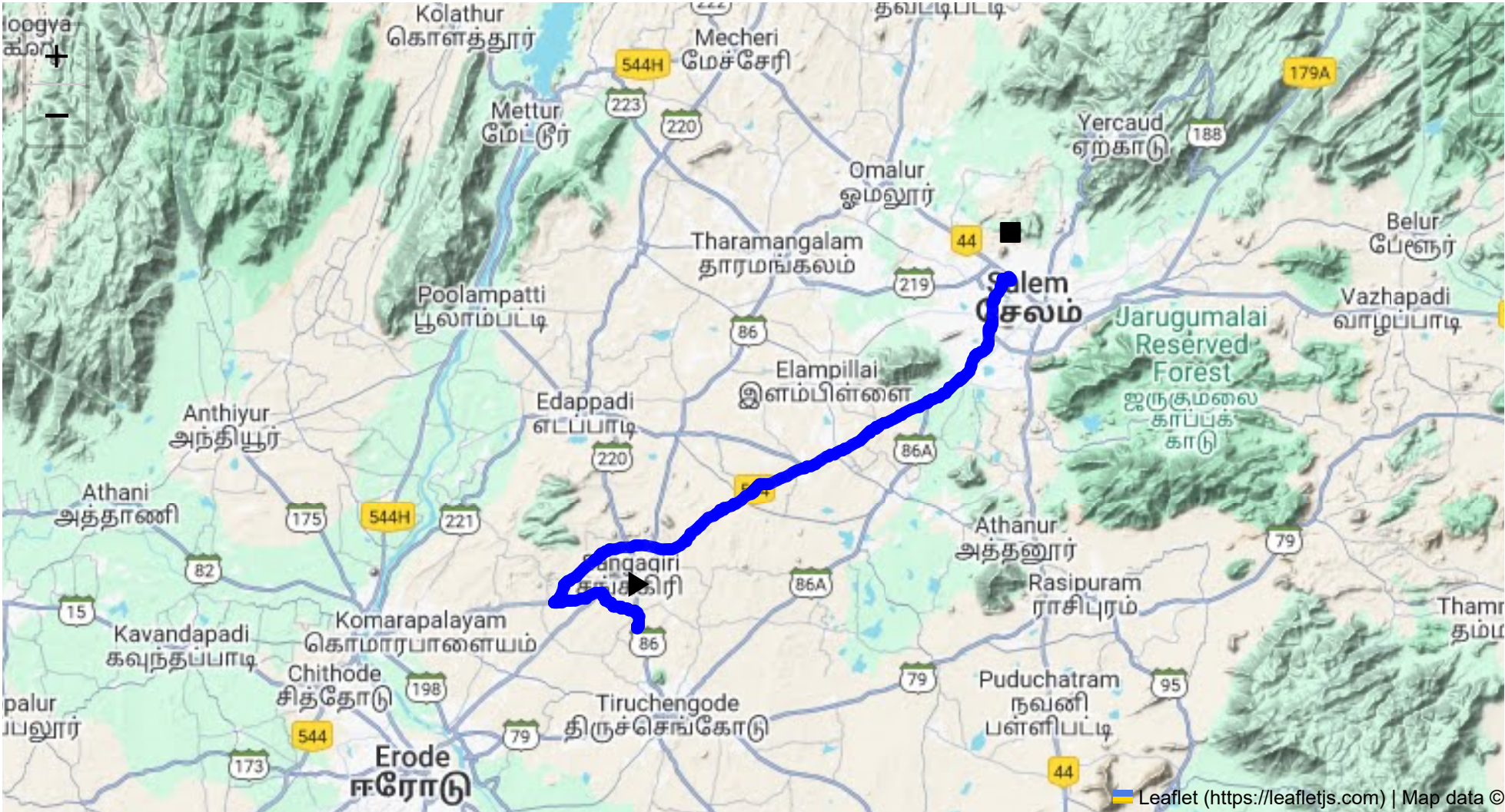
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 54.57 km
Estimated Duration: 1.1 hours
Adjusted Duration (Heavy Vehicle): 1.3 hours
Start: (11.4381, 77.8734)
End: (11.677671, 78.132439)



Welcome to the Journey Risk Management Study

Route Overview

The route from CVQF+23W, Sangagiri to M4HM+22R, Swarnapuri Annexe in Salem spans approximately 54.57 kilometers. It passes through key points such as FV2C+2PH (Sankari RS - Erode Privu Rd), FR7X+844 (Bhavani Main Road, Sankari West), and FR48+7XQ (Bhavani Main Road, Padaiveedu). This route typically takes about 1.06 hours for heavy vehicles carrying hazardous materials.

Weather Conditions and Hazards

The region experiences a tropical climate with hot, dry summers and wet monsoons. The monsoon season from June to September can cause heavy rainfall leading to waterlogging, reduced visibility, and slippery road conditions. Occasional thunderstorms may occur, particularly during the pre-monsoon and monsoon months.

Traffic Patterns

This route can experience congestion, particularly at junctions near towns and during peak hours: morning (8 AM - 10 AM) and evening (5 PM - 8 PM). The roads near Sankari and Padaiveedu are prone to higher traffic density.

Road Quality and Infrastructure

The road quality varies, with some sections prone to potholes, particularly after the monsoon season. Key junctions and major roads generally have better maintenance, while rural segments may lack proper markings and lighting.

Alternative Routes

In emergencies, consider detouring via NH44 which provides a more direct route between Sankagiri and Salem, albeit with higher traffic. Another alternative route is via the Tiruchengode to Sankagiri Road and then connecting back to NH44.

Local Regulations on Hazardous Materials

Transport of hazardous materials is subject to strict regulations. Ensure adherence to the Central Motor Vehicle Rules, 1989, which require proper documentation, labeling, and equipment for handling accidental spills.

Historical Incidents

The area has a relatively low incidence of accidents involving heavy vehicles and hazardous materials. However, occasional reports of overturning trucks and road mishaps do occur, particularly during adverse weather conditions.

Environmental Considerations

The route passes through semi-urban and rural landscapes with agricultural lands. Ensure measures to prevent spillage in these sensitive areas to protect both ecosystems and local water sources.

Communication Coverage

Communication coverage along this route is generally good, with occasional dead zones in rural sections. However, major telecom providers cover the area, though signal strength may fluctuate.

Emergency Response

Emergency response times can vary, but hospitals and facilities in Salem are well-equipped. Roadside assistance may take longer in rural stretches, potentially exceeding 30 minutes in remote areas.

Overall Risk Assessment

The route is moderately safe for transporting hazardous materials with adequate precautions. Key concerns include weather conditions, road quality in rural areas, and peak traffic congestion. Awareness of local regulations and emergency contacts is essential. Continuous monitoring of weather forecasts and real-time traffic updates can mitigate risks effectively. Careful driving practices and route familiarization are recommended for safer transport.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.67563, 78.13375	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43956, 77.87340	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Turn	High	11.44029, 77.87544	15 KM/Hr
5	Turn	Medium	11.45357, 77.85789	30 KM/Hr
6	Turn	Medium	11.45352, 77.85698	30 KM/Hr
7	Turn	Medium	11.46303, 77.84945	30 KM/Hr
8	Turn	Medium	11.46318, 77.84913	30 KM/Hr
9	Turn	High	11.45542, 77.81725	15 KM/Hr
10	Turn	High	11.45631, 77.81668	15 KM/Hr
11	Turn	Medium	11.61736, 78.10588	30 KM/Hr
12	Turn	Medium	11.61753, 78.10608	30 KM/Hr
13	Turn	Medium	11.64007, 78.11794	30 KM/Hr
14	Turn	Medium	11.64020, 78.11786	30 KM/Hr
15	Turn	High	11.67074, 78.12668	15 KM/Hr
16	Turn	High	11.67081, 78.12661	15 KM/Hr
17	Turn	High	11.67582, 78.12786	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
4	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
5	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
6	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
9	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
13	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
14	hospital	Sisu Hospital Inida Ltd	11.6505276, 78.1205003	30 km/h	Medium
15	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
16	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium
17	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
18	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium
20	hospital	Venkatesh Hospital and Urology Centre	11.6759517, 78.1324029	30 km/h	Medium
21	hospital	R.K. Hospital, Salem	11.6746905, 78.1339717	30 km/h	Medium
22	hospital	Kurinji Hospital, Salem	11.6771806, 78.1351224	30 km/h	Medium
23	hospital	Bhavani Hospital	11.6745697, 78.1362683	30 km/h	Medium
24	clinic	Ayush Care Siddha Centre	11.6731195, 78.1351967	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
25	clinic	Bismillah Piles Clinic	11.6755285, 78.1343325	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
8	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
10	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
11	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
12	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
19	college	Sona College	11.6768, 78.1257052	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High

Speed Limit: 15 KM/Hr
Coordinates: 11.67563, 78.13375



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45357, 77.85789



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46318, 77.84913



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45631, 77.81668



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61736, 78.10588



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61753, 78.10608



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64007, 78.11794



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64020, 78.11786



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.67074, 78.12668



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67081, 78.12661



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67582, 78.12786

